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# Algorithmic Pathways to Mental Health Content: How YouTube's Recommendation System Shapes Adolescent Exposure Patterns

## Abstract
Pew Research survey in 2022 showed that 95% of US adolescent teens (ages 13 to 17) have used YouTube, and 34% of teens get mental health info on social media. YouTube, despite being a massive influence as a source of teen's mental health resource we yet lack systematic understanding of how its recommendation algorithm shapes help-seeking users' content exposure. This study examined YouTube's mental health content curation through six algorithmic walkthroughs totaling 201 manually analyzed videos and validated with 525 videos collected via API across 12 search terms. The findings of this study revealed three critical failures: (1) pathway bifurcation determined by search framing (negative/clinical vs. positive/wellness) and channel engagement (established ecosystem vs random), creating a "danger zone" where vulnerable engagement where vulnerable searches using problem-focused language encountered gradual escalation from general advice to crisis/extreme content without professional intervention; (2) systematic algorithmic suppression of professional mental health resources, which despite constituting 24.4% of available content and achieving 32.5% higher engagement than peer content, appeared in fewer than 10% of typical user pathways; and (3) catastrophic disadvantage for teen-focused content, showing only 8% availability, with engagement rates 50% below the average, and view counts 84% below the mean. A dual-axis framework predicting pathway outcomes based on search characteristics and channel engagement achieved perfect classification (6/6) journey and identified specific intervention points. Results demonstrate that YouTube's algorithm functions as a sentiment amplifier rather than neutral information channel, systematically exposing the most vulnerable user groups to the most harmful/dangerous content while suppressing high-performing professional resources. The platform's governance needs to evolve beyond content moderation to recommendation accountability, ensuring algorithms fulfill responsibility for adolescents seeking mental health support.

**Keywords:** YouTube, recommendation algorithms, adolescent mental health, platform governance, algorithmic accountability, mental health content, social media

## Introduction
When a teenager searches YouTube for "depression help", they initiate a journey shaped not by clinical expertise but by the algorithm's curation. More than 95% of U.S. adolescents use YouTube (Anderson & Jiang, 2018), and many turn to the video sharing platform during mental health crises, seeking support, understanding, and even connect/relate to others who share their struggles. Despite YouTube's overwhelming influence as a mental health resource for young people, we have minimal knowledge about what kind of content these people actually encounter.

The answer is crucial: adolescent depression and anxiety have reached crisis levels in recent years (Twenge et al., 2019), with rates of major depressive episodes among teens increasing 52% between 2005 and 2017. For many vulnerable teens and adolescents, YouTube's recommendation serves as their first- and in some cases their only guide to mental health recommendation. The gap in understanding is particularly concerning given evidence that recommendation algorithms can systematically amplify extreme content in other domains (Ribeiro et al., 2020). If curated algorithms can drive users toward political radicalization, what possible pathways can it lead to for teens who are seeking mental health support? This study addresses this very question through systemic algorithmic simulation to understand adolescent help-seeking behaviors on YouTube.

Research on algorithmic recommendation systems reveals that these platforms are not neutral sources of information. Ribeiro et al. (2020) demonstrated that YouTube's recommendation algorithm systematically guides users toward increasingly extreme political content through a process of progressive radicalization pathways. Similarly, scholars have documented how social media platforms amplify filter bubbles (Yeung et al., 2022) and enable the spread of health misinformation (Yeung et al., 2022). Yet despite extensive research on algorithmic amplification in political and commercial contexts, we lack systematic understanding of how recommendation algorithms shape adolescent exposure to mental health content, a crucial domain where algorithmic failures could have life or death consequences for vulnerable youth who are seeking mental health advice through these channels.

The results of algorithmic curation become quite severe when applied to mental health content, where adolescents increasingly turn to digital platforms for support as it has become easily accessible. Research shows that peer-created mental health content dominates online spaces, with young people sharing personal narratives of depression and anxiety (Nesi et al., 2021). Similarly, adolescents with mental illness and depressive symptoms were more likely to use social media platforms to interact with users (Nesi et al., 2021). Even though such peer content can provide authenticity and avoid stigma, studies have shown concerning patterns like Instagram hosts thriving pro-eating disorder communities using coded hashtags to get around content moderation (Chancellor et al., 2016), TikTok amplifying mental health self-diagnosis videos with questionable accuracy (Yeung et al, 2022), and Youtube featuring both evidence based professional resources and potential harmful peer-narratives side by side. However this research has predominantly focused on the type of content that exists rather than understanding what adolescents actually encounter through algorithmic recommendations. The distinction is crucial; content availability does not ensure content exposure. A platform might host great professional mental health resources but if the recommendation algorithm systematically refers to peer content or entertainment, those resources remain invisible to teens who are seeking critical help. While scholars and studies have documented the diversity of mental health related content online, there is a systematic lack of understanding of how YouTube's recommendation algorithm curates adolescent exposure in terms of which resources are amplified, which are suppressed, and what journey the pathways for such content lead to for teens who are seeking help.

Despite growing recognition that algorithms shape user experience and that mental health content varies in terms of quality dramatically, these two research streams still remain

disconnected. We understand that recommendation algorithms can systematically amplify extreme content, driving users towards political radicalization (Ribeiro et al., 2020) or even ideological filter bubbles (Bakshy et al., 2015). We also know mental health content that is available online ranges from professional resources to questionable peer advice to vulnerable individuals (Chancellor et al., 2016; Yeung et al., 2022). However, we lack systematic understanding of how these two dynamics intersect: when vulnerable help seeking adolescents seek out depression help on YouTube, what does the algorithm actually surface? Do different search phrases impact content quality or differ content quality? Does the algorithm simply amplify professional mental health resources or suppress them in favor of engaging peer content? What factors determine whether a help-seeking search leads to professional support or harmful advice? This study addresses these questions through systematic algorithmic audits simulating adolescent mental health searches and map the recommendation pathways following the journey that it leads.

This research applied algorithmic walkthrough methodology, conducting six systematic journeys that each began with a different mental health search query and followed YouTube's top video recommendations for approximately 20-90 videos. Fresh incognito browser sessions were used to prevent algorithmic cross contamination where I manually analyzed 201 videos across various topics ranging from clinical terms ("mental health", "teen depression") to emotional language ("feeling depressed") to help seeking ("therapy for teens") and wellness-focused searches ("mental wellness"). To validate these findings at scale, I collected additional 525 videos through YouTube's API across mental health search terms, enabling quantitative analysis of channel ecosystem, professional content and engagement patterns among these channels.

The findings reveal systematic algorithmic failures that disproportionately cause harm to the most vulnerable teens. When a search was initiated for "mental health", it began with a random life coach video, and started escalating over 90 videos from general wellbeing advice to increasingly darker content from loneliness to hopelessness to videos discussing suicidal thoughts; in a journey where professional mental health resources appeared in fewer than 2% of recommendations despite constituting 24.4% of YouTube's mental health content corpus. Similarly, searches which used vulnerable emotional language led to the least helpful outcomes: "feeling depressed" triggered a loop where the algorithm pushed depression-themed films rather than therapeutic resources, with around 22.2% of content which could be potentially harmful. Likewise, help-seeking language with age specification revealed audience misclassification: "therapy for teens" - a likely search made by teens seeking help for themselves resulted in parent-focused content about managing their teen children's mental health which positioned teens as objects of intervention rather than subjects who are actively seeking help.

These divergent pathways revealed systematic patterns that were captured in a dual-axis framework: the algorithm creates pathway bifurcation that is determined by search term framing (negative/clinical vs positive/wellness) and channel engagement (established mental health channels vs random channels). Searches with positive framing had algorithmic protection even if the journey started with a random unestablished channel. Searches with "mental wellness" maintained helpful self-care content throughout the journey while negative framing with established channels created safe but problematic loops from audience mismatch, topic drift or worse single channel loop. When validated with large-scale API validation, it was confirmed that

there was systematic algorithmic suppression: professional mental health content, despite showing 32.5% higher engagement rates compared to peer content, appeared fewer than 10% of typical user pathways. Alarmingly, teen-specific searches resulted catastrophic outcomes where only 8% professional content was available for "teen mental health", engagement rates was 50% below average and view counts 84% below dataset mean - representing platform's failure to serve adolescent help-seekers.

This research contributes three findings: first, a dual-axis framework that predicts whether search behavior leads to helpful or harmful algorithmic journey pathways; second: a taxonomy of six distinct recommendation patterns ranging harmful/dangerous escalation to protective loops; and third, identification of dual failure mode requiring intervention at both algorithmic curation (professional content exists but doesn't surface) and content creation (teen-specific resources unavailable/underproduced) levels.

## Methods

### Research Design

This study used a sequential mixed-methods design combining qualitative algorithmic walkthrough with quantitative API data validation. The research was conducted in two phases: first systematic navigation and analysis of YouTube's recommendation pathways to understand user experience from scratch; second phase involved large-scale data collection to validate emergent patterns and quantify algorithmic behaviors observed qualitatively. This design enabled both deep contextual understanding of a user's journey (teens) and statistical generalization ability across the broader YouTube ecosystem.

The analytical approach followed grounded theory principles (Glaser & Strauss, 1967), allowing theoretical frameworks to emerge from data rather than imposing predetermined categories. In this study, I began with an open observation of recommendation patterns across diverse mental health searches, iteratively identifying common patterns and differences that coalesced into the dual-axis framework and loop taxonomy that are presented in findings. Through quantitative validation which was later tested for qualitative patterns held at scale, whether professional content suppression was systematic, and whether teen content disadvantage persisted across the full corpus.

This mixed-method approach addresses the limitations inherent in either of those methods by itself. Algorithmic walkthrough captures nuanced user experience - progressive escalation of content, sentimental shifts, turning points - that collective stats obscure; API data collection provides statistical power and prevents against researcher bias through validation via qualitative observations with platform-wide patterns. This joint approach offers complementary insights into how the recommendation algorithm shapes adolescent mental health content exposure.

### Phase 1: Algorithmic Walkthrough

Algorithmic walkthrough methodology enables systematic study of recommendation systems from a user's perspective, documenting the pathways that emerge from specific starting points (Bucher, 2018). Instead of analyzing algorithms from outside through code inspection (unable for proprietary platforms like YouTube) or solely through aggregate API data (which undermines

individual user experience), this approach navigates recommendation pathways firsthand to understand what teens actually encounter.

**Journey Protocol**

Each journey in this study began with a fresh incognito browser session to prevent algorithmic personalization based on viewing patterns and history, simulating a new user's experience. I navigated to YouTube.com, entered a specific mental health search query, and examined the initial search results without logging in. From the search results, I selected and watched the first video for approximately 2-3 minutes to assess content type, creator credentials, sentiment, and topic. This viewing behavior mirrors realistic adolescent attention patterns which is sufficient to determine whether the content is relevant and helpful without requiring full video consumption. After viewing the first video, I examined YouTube's sidebar recommendation videos and selected one of the top two-three videos (typically positioned 1-2), simulating typical user clicking behavior where the most engagement occurs with top recommendations (Zhou et al., 2010). This process was repeated: watching the video, selecting top recommendations for the next video until reaching a natural stopping point.

**Stopping Criteria**

The algorithmic journeys were concluded when one of the four conditions occurred: (1) algorithmic loop formation, where the same videos or channels appeared repeatedly in recommendations, indicating pathway stabilization; (2) pattern saturation, where recommendation patterns became predictable and stopped revealing new information; (3) natural endpoints, such as reaching professional crisis resources or harmful content requiring intervention; or (4) substantial journey length (80-90 videos), providing enough data to generalize the pathway. The journey length in this study ranged from 20 videos (when loops were formed quickly) to 90 videos (when exploration continued without stabilization).

**Search Term Selection**

There were six search terms which were chosen to represent a range of diverse adolescent help-seeking behaviors, differing on three dimensions: clinical versus emotional language, specificity versus generality, and problem-focused wellness-focused framing. The terms included: (1) "mental health" (general, clinical, neutral), (2) "feeling depressed" (specific, emotional, negative), (3) "therapy for teens" (help-seeking, age-specific), (4) "anxiety relief" (solution-focused, positive framing"), (5) "teen depression" (age-specific, clinical), and (6) "mental wellness" (general, positive, proactive). This variation showed comparisons of how search phrasing affects the outcome of the algorithm - whether vulnerable emotional language produced different recommendations than clinical terms, whether age-specific searches surface age-appropriate content, and whether positive framing creates protective pathways.

**Data Documentation**

For every video encountered, I recorded: video ID and video URL, content type, sentiment, target audience, resource quality, has professional references, potential of harm, the date it was coded, journey number, position in journey, starting condition, creator type, and production quality. Qualitative observations included: content type classification credentials (licensed therapist/psychologist, peer with lived explainer, entertainment, commercial), creator sentiment and tone (positive/hopeful, neutral/informational, negative/despairing), presence of professional

resources (crisis hotlines, therapy referrals, evidence-based coping strategies), potentially harmful elements (romanticization of mental illness, triggering content, misinformation). I also documented recommendation patterns, noting when loops formed, when content shifted in type or sentiment, and when professional resources appeared or disappeared from recommendations.

**Sample**

Six journeys produced 201 manually analyzed videos: Journey 1 ("mental health") yielded 90 videos following gradual escalation; Journey 2 ("feeling depressed") reached 20 videos before stabilizing in a content-type loop; Journey 3 ("therapy for teens") analyzed 30 videos of audience-mismatched content; Journey 4 ("anxiety relief") examined 20 videos dominated by single-source professional content; Journey 5 ("teen depression") documented 20 videos exhibiting topic drift; and Journey 6 ("mental wellness") followed 30 videos maintaining positive framing. This corpus represents the recommendation pathways accessible to adolescents using common mental health search terms on Youtube as of January 2026.

**Limitations of Method**

API data collection used in this study captures available content but not actual recommendation sequences - while these videos exist in YouTube's corpus, API results don't ensure they appear in user recommendations. The binary professional/peer classification may obscure important distinctions (example: experienced peer advocates versus unqualified social media influencers). Also, channel credentials relied on self-reported information, which might contain inaccuracies. Finally, the 525 video sample, while substantial, represents a fraction of YouTube's total mental health content, there's a high probability that niche communities or less-popular high quality content might be actually missing.

Table 1 summarizes journey characteristics:

| Journey | Search Term | Videos | Starting Channel Type | Outcome Pattern |
|---|---|---|---|---|
| 1 | "mental health" | 90 | Random (life coach) | Gradual escalation |
| 2 | "feeling depressed" | 20 | Established (therapist) | Content-type loop (films) |
| 3 | "therapy for teens" | 30 | Established (therapists) | Audience mismatch |
| 4 | "anxiety relief" | 20 | Established (doctor) | Single source dominance |
| 5 | "teen depression" | 20 | Established (doctor) | Topic drift |
| 6 | "mental wellness" | 30 | Random (wellness creator) | Positive overall |

“This methodology builds on prior platform studies examining YouTube radicalization (Ribeiro et al., 2020), TikTok algorithmic assumptions (Klug et al., 2021), and Instagram content moderation (Chancellor et al., 2016), adapting their observational approaches to mental health content pathways."

**Ethical Considerations**
This research only analyzed publicly available content and did not collect data on individual users, protecting privacy. All observations took place in researcher-controlled browser sessions without tracking other user’s behavior. This study was conducted as platform observation rather than human subjects research, with the purpose of examining recommendation system behavior rather than individual user responses. Viewing potentially harmful content (suicidal thoughts, mental crisis) was limited to research documentation purposes with researcher self-care protocols including breaks between journeys and debriefing.

**Phase 2:**

**API Data Collection**
To validate qualitative journey findings at scale and quantify patterns observed through algorithmic walkthrough, I collected a broader corpus of mental health content using YouTube’s Data API v3. This large scale collection enabled statistical analysis of channel ecosystems, professional content distribution, and engagement metrics across the mental health content landscape.

**Sampling Strategy**
I queried YouTube’s API with 12 mental health-related search terms, including the six terms which were used in algorithmic walkthroughs with six additional variations: “depression help”, “depression support”, “anxiety support”, “coping with anxiety”, “self care mental health”, and “how to be happy”. These additional terms provided a broader scope of help-seeking language (e.g. “depression help”), support-seeking (e.g. “depression support”, “anxiety support”), and wellness-focused searches (e.g. “self care mental health”). For every search term, I retrieved 25-50 videos using the API’s search endpoint with parameters set to: relevance-based ranking (YouTube’s default recommendation logic), English language content, video-only results (excluding playlists and channels), and no geographic restrictions. This produced a corpus of 525 videos which were collected in January 2026.

**Variables Collected**
For each video, the API returned: video ID, title, description channel name and ID, publication date, thumbnail URL, tags, category ID, duration, view count, like count, and comment count. I calculated engagement rate as (likes + comments)/views * 100 providing a normalized metric for comparing content performance across videos with different view counts. Collection date was recorded to enable temporal analysis if algorithms changed.

**Channel Ecosystem Analysis**
In order to quantify the “established ecosystem” observed qualitatively in journeys, I identified channels which appeared frequently across multiple search terms. Operational criteria defined

“established channels” as those which met two thresholds: (1) at least 10 videos in the collected corpus, indicating substantial mental health content production, and (2) appearance in at least 5 different search terms, demonstrating cross-topic relevance rather than narrow specialization. This analysis identified 8 channels which accounted for 36.4% of the dataset confirming the existence of an algorithmically-previleged channel ecosystem.

**Professional Content Classification**

The videos in this study were classified as “professional” if created by channels operated by licensed mental health professionals (therapists, psychologists, psychiatrists, clinical social workers) or credentialed researchers (neuroscientists, medical doctors specializing in mental health). Channel credentials were verified through: channel “About” pages listing professional credentials, video content indicating clinical training, organizational affiliations with mental health institutions, and cross-referencing with professional directories where available. Some of the identified professional channels included Therapy in a Nutshell (licensed therapist), Dr. Tracey Marks (psychiatrist), Dr. Julie (clinical psychologist), The Grateful Therapist (licensed therapist), Andrew Huberman (neuroscientist), HealthyGamerGG (psychiatrist), and other licensed professionals. Content from non-licensed creators (life coaches, peer advocates, motivational speakers, organizations without clinical staff) were classified as “peer/other” regardless of content quality. This binary classification approach allowed statistical comparison of how algorithms surface professional versus peer-created mental health content.

**Data Validation**

In order to ensure API data reflected actual recommendation behavior rather than search-only results, I cross-referenced API-collected videos with videos encountered in algorithmic walkthroughs. High overlap confirmed that API search results approximated the content pool from which YouTube’s recommendation algorithm draws. In addition, I examined whether the top channels in the API data matched with the channels frequently appearing in journey recommendations, leading to strong correspondence (e.g. Psych2Go, Tedx Talks, Therapy in a Nutshell appeared prominently in both datasets).

**Limitations**

API data collection captures available content but not the actual recommendation sequences - while these videos exist in YouTube’s corpus. API results necessarily don’t guarantee they appear in user facing recommendations. The binary approach to professional/peer classification might obscure important distinctions (e.g., experienced peer advocates versus unqualified influencers). Finally the 525-video sample, while being substantial, represents only a small fraction of YouTube’s total mental health content that’s available, potentially missing niche communities or even less-popular but high-quality resources.

**API Implementation**

The data collection process used Python 3.9 with the ‘google-api-python-client’ library (v2.108.0). The search endpoint (‘youtube.search().list’) retrieved video IDs, which were then passed to the videos endpoint (‘youtube.videos().list’) to gather detailed statistics and metadata. Rate limiting was implemented (0.3-second delay between requests) to comply with YouTube’s API usage policies. All data was stored in CSV format for subsequent analysis using pandas (v2.1.0) for data manipulation and matplotlib/seaborn for visualization.

Table 2 shows the rationale for additional search terms beyond the six used in algorithmic walkthroughs:

| Search Term | Rationale | Expected Content |
|---|---|---|
| “depression help” | Direct help seeking language | Professional resources |
| “depression support” | Support-seeking (community) | Peer support groups |
| “anxiety support” | Support-seeking | Peer + professional mix |
| “coping with anxiety” | Action oriented strategy | Coping techniques |
| “self care mental health” | Wellness maintenance | Self-help strategies |
| “how to be happy” | Positive outcome focused | Motivational content |

The complete list of professional channels include: Therapy in a Nutshell (Emma McAdam, LMFT), Dr. Tracey Marks (psychiatrist), Dr. Julie (clinical psychologist), The Grateful Therapist (licensed therapist), Andrew Huberman (Stanford neuroscientist), HealthyGamerGG (Dr. Alok Kanojia, psychiatrist), MedCircle (mental health platform with licensed clinicians), Psych Hub (educational organization with clinical advisors), and Doc Snipes (Dr. Dawn-Elisse Snipes, LPC-MHSP). These channels collectively produced 128 videos (24.4%) of the 525-video corpus.

**Analytical Approach**

**Qualitative Analysis**

Journey data were analyzed using grounded theory principles (Glaser & Strauss, 1967; Charmaz, 2006), allowing patterns and theoretical frameworks to emerge from systematic comparison across cases rather than imposing predetermined categories. The analysis was conducted iteratively: initial open coding identified content types, creator credentials, sentiment analysis, and recommendation behaviors across all 201 videos. Axis coding then grouped observations into order patterns, it revealed journey diverged based on two key dimensions - search term framing and channel engagement type. Selective coding helped refining these dimensions into the dual-axis framework, with search framing (negative/clinical vs. positive/wellness) forming one axis and channel type (established ecosystem vs. random forming the other.

Loop patterns emerged through comparative analysis of when and how journeys were stabilized. Journey 2 had repetitive awareness film recommendations exemplified content-type loops whereas Journey 3 had consistent parent-focused content despite teen-directed search terms which revealed audience mismatch. Journey 4 illustrated single-channel dominance and creator amplification and Journey 5 drifted from depression to attachment theory showing a massive topic drift. Journey 6 maintained wellness focus despite starting with a random channel and showed positive framing protection. Journey 1, unlike others, saw a progressive escalation without stabilization representing absence of loop formation, and was classified as dangerous escalation. These six patterns constituted the loop taxonomy.

## Quantitative Analysis

The API data analysis focused on three main objectives: validating the established ecosystem, comparison between professional and peer content performance, and examining search term variation. Channel ecosystem variation calculated video counts and search term diversity for all channels, identifying those meeting established ecosystem criteria (10+ videos, 5+ search terms). This quantitative confirmation supported qualitative observations that certain channels dominated recommendations across diverse mental health searches.

Search term analysis examined how professional content availability, average engagement, and average views varied across the 12 queries. There were revealed systematic patterns: help-seeking language ("depression help") yielded 56% professional content while teen-specific searches ("teen mental health" produced only 8%; positively-framed searches ("how to be happy") achieved highest engagement (4.80%) while teen-focused queries returned the lowest (1.78%). These distributions quantitatively validated journey-based observations about how search phrasing shapes content exposure and recommendations.

## Triangulation

Findings were triangulated across methods to ensure robustness. For example, observations in Journey 1 of rare occurrences of professional content was validated by API data where it showed only 14% professional availability for "mental health" searches and less than 25 appearances in the actual journey pathway. Likewise, Journey 3's audience mismatch finding was corroborated by API data "therapy for teens" content averaged 198,526 views (lowest found in dataset), suggesting poor engagement potentially due to audience mismatch. Journey 4's single channel dominance aligned with API data showing that while "anxiety relief" searches contained only 10% of professional content overall, those few professionals achieved high visibility. This

convergence of qualitative observation and quantitative measurement strengthened the confidence in findings.

**Framework Validation**
The dual-axis framework's predictive validity was assessed by testing whether all six journeys fit their predicted quadrants based on search framing and channel engagement. In Journey 1 (negative framing + random channel) occupied the predicted danger zone with little-no professional help. Journey 2, 3, and 5 (negative framing + established channels) fell into predicted safe but flawed loops. Journey 4 (positive framing + established channel) matched the professional content quadrant, and Journey 6 (positive framing + random channel) aligned with the positive loop quadrant. Perfect classification (6/6) supported the framework's explanatory power and potential generalizing ability to untested search terms.

As for statistical analysis, it used Python's scipy.stats library. Independent samples t-tests compared professional versus peer content on engagement metrics. Effect sizes were calculated using Cohen's d. For non-normally distributed variables (view counts that exhibited right skew), Mann-Whitney U-tests supplemented t-tests. Significance threshold was set at $\alpha = 0.05$. All tests were two-tailed.

**Reliability Considerations**
As this was a single-coded study, inter-rater reliability could not be calculated. In order to mitigate subjectivity, I employed explicit coding criteria: professional content required verifiable credentials (licenses, degrees, institutional affiliations), content-type classifications used mutually exclusive categories with decision rules (e.g. "educational" required explanatory focus, "peer narrative" required first-person story telling), and sentiment coding distinguished emotional tone(positive/neutral/negative) with informational quality. Future research should employ multiple coders to quantify consensus on these classifications.

**Results**

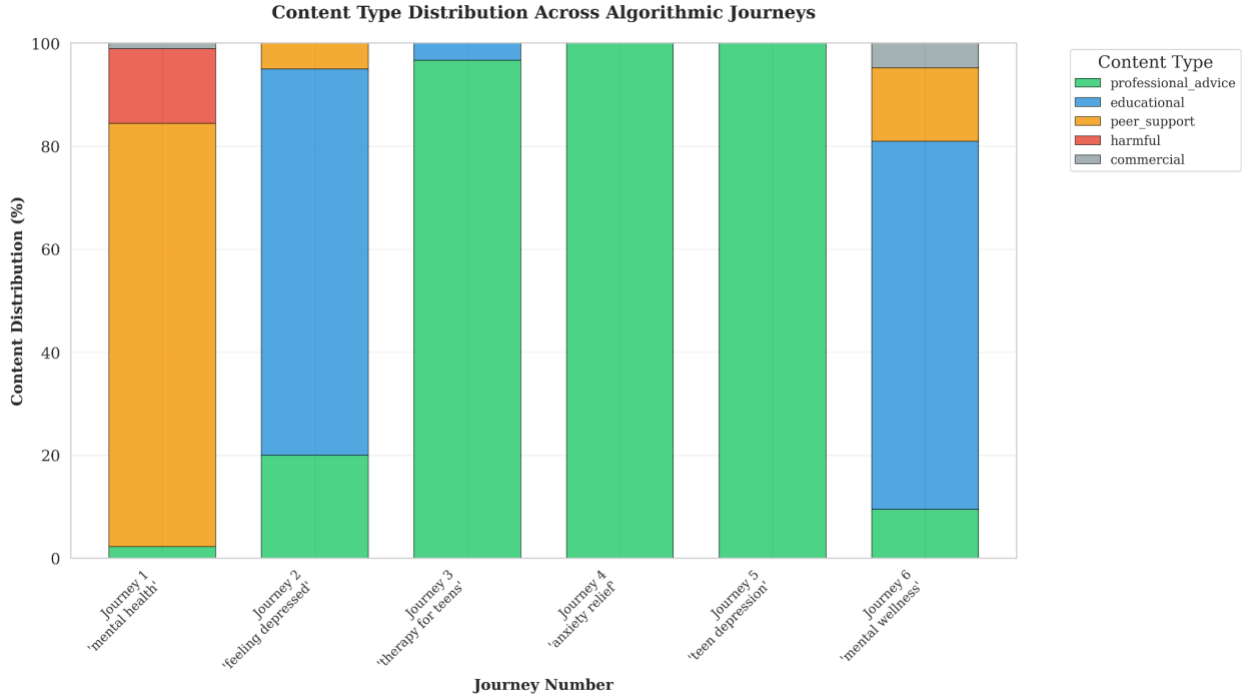
**Overview of Algorithmic Pathways**
Six algorithmic journeys produced 201 manually-analyzed videos, revealing systematic variation in recommendation pathways based on search query characteristics and user engagement patterns. Journey length ranged from 20 videos (when algorithmic loops were formed rapidly) to 90 videos (when pathways continued exploring without stabilization), with a mean 33.5 videos per journey (SD = 28.7). Table 3 summarizes journey characteristics:

**Table 3: Summary of Algorithmic Journey Characteristics**

| Journey | Search Term | Videos | Starting Channel | Channel Type | Outcome Pattern | Professional Content | Harmful Content |
|---|---|---|---|---|---|---|---|
| 1 | "mental health" | 90 | Life coach | random | Progressive escalation | 2.2% (2 video) | 22.2% (20 videos) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | “feeling depressed” | 20 | Therapist | Established | Content-type loop | 95.0 % (19 videos) | 5% (1 videos) |
| 3 | “therapy for teens” | 30 | Multiple therapists | Established | Audience mismatch | 100 % (30 videos) | 0% (0 videos) |
| 4 | “anxiety relief” | 20 | Doctor (Doc Snipes) | Established | Single-source loop | 100% (20 videos) | 0% (0 videos) |
| 5 | “teen depression” | 20 | Doctor (children’s institute”) | Established | Topic drift loop | 100% (20 videos) | 0% (0 videos) |
| 6 | “mental wellness” | 21 | Wellness creator | Random | Positive maintenance | 81% (17 videos) | 0% (0 videos) |

**Note:** Professional content = videos created by licensed mental health professionals (therapists, psychologists, psychiatrists, neuroscientists etc.). Harmful content = videos containing potentially triggering material, romanticization of mental illness, or crisis level content without resources. Channel type classification based on whether the starting channel belonged to the established ecosystem (8 dominant channels identified through API analysis).




Across all journeys, professional content appeared in 53.7% of videos (108 of 201), but this aggregate masked dramatic variation across journeys. Journeys 3, 4, and 5 featured nearly exclusive professional content (96-100%), while Journeys 1 and 2 showed minimal professional

presence (1-5%). Potentially harmful content appeared in 9.5% of all videos (19 of 201), concentrated heavily in Journeys 1 and 2 (both used negative/clinical search framing with different starting channels). These differences in outcome quality despite similar mental health topics suggest systematic algorithmic differentiation based on search and engagement characteristics, motivating development of the dual-axis framework.

The established ecosystem- 8 channels identified through API analysis appeared over 5+ search terms with 10+ videos each (TEDx Talks, Psych2Go, Therapy in a Nutshell, Dr. Tracey Marks, Dr Julie, Mel Robbins, HealthyGamerGG, Andrew Huberman) - dominated Journey 3, 4, and 5, where starting channel engagement or early clicks intersected with these creators. In contrast, Journeys 1 and 6 began outside this ecosystem with random creators (life coach, wellness influencer), producing divergent outcomes: escalation to harmful content (Journey 1) versus maintained positive framing (Journey 6). This pattern suggested channel type alone did not determine outcomes; search framing also played a critical role.

Search Terms varied across three dimensions: emotional versus clinical language ("feeling depressed" vs "mental health"), age-specific versus general ("teen depression" vs "depression"), and problem-focused versus well-oriented ("anxiety relief" vs "mental wellness"). This variation enabled systematic comparison of how search phrasing affects algorithmic outcomes.

## Results

### The Dual-Axis Framework

Systematic comparison across the six journeys revealed that algorithmic outcomes were determined by two independent factors operating in combination: search term framing and channel type. These factors formed orthogonal axes creating four distinct outcome quadrants, each associated with specific recommendation patterns and content quality outcomes.

### Axis 1: Search Term Framing

The first axis differentiated between negative/clinical framing and positive/wellness framing in search queries. Negative or clinical searches employed problem-focused language referring to mental health conditions or emotional distress ("mental health", "feeling depressed", "teen depression"). On the contrary, positive or wellness searches used solution-oriented or proactive language with emphasis on wellbeing and coping mechanisms ("mental wellness", "anxiety relief"). This linguistic distinction proved algorithmically consequential distinction: negative framing led the algorithm to amplify content about mental health problems, depression narratives, and crisis-level material, while positive framing recommendations were towards self-care strategies, motivation, and wellness maintenance. The framing effect operated independently of content availability even when professional resources existed for negatively-framed searches, the algorithm systematically surfaced more dramatic or emotionally resonating peer content instead.

### Axis 2: Channel Engagement Type

The second axis distinguished between engagement with the established ecosystem versus random channels. The established ecosystem had eight channels identified through API analysis as dominating mental health content across multiple search terms: TEDx Talks, Psych2Go,

Therapy in a Nutshell, Dr. Tracey Marks, Dr Julie, Mel Robbins, HealthyGamerGG, and Andrew Huberman. These channels collectively accounted for 36.4% of all API-collected videos and appeared consistently across diverse mental health searches. Interacting with these channels either by initiating searches that surfaced them or clicking their content early in a journey triggered algorithmic pathway constraints which kept subsequent recommendations within this bounded ecosystem. On the contrary, engaging with channels outside the ecosystem (random life coaches, small wellness creators, individual peer advocates) enabled broader algorithmic exploration without inbuilt safety guardrails allowing the recommendations to drift towards more extreme or varied content based on engagement signals alone.

**Four Quadrants, Four Outcomes**

The intersection of these axes produced four-distinct pathway types (Figure 2). Quadrant 1 (Negative framing + Random channels) represented "Danger Zone": algorithmic exploration without safety constraints, amplifying negative sentiment progressively. Journey 1 demonstrated this quadrant which escalated from general mental health advice to suicidal thoughts over the course of 90 videos. Quadrant 2 (Negative framing + Established channels) produced "Safe but Flawed Loops": the established ecosystem prevented extreme content escalation, but algorithmic failures persisted through content type mismatches (short films instead of resources), audience misidentification (parent content for teen searches), or topic drift (related but non-responsive content). Journeys 2, 3, and 5 fall in this quadrant. Quadrant 3 (Positive framing + Established channels) resulted in "Professional Content": engagement with established channels combined with solution seeking language produced consistent professional resources often through single source domination. Journey 4 demonstrated this pattern with 100% professional content but limited to a single doctor's channel. Quadrant 4 (Positive framing + Random Channels) created so-called "Positive Loops" which despite random channel starts, positive wellness framing maintained helpful self care content without escalation. Journey 6 exemplified this protective effect.

**Framework Validation**

The framework achieved perfect classification: all six journeys fit their predicted quadrants based on their search framing and initial channel engagement. The predictive accuracy suggested the dual-axis model captured core algorithmic logic determining recommendation pathways. The framework also explained the counterintuitive finding that Journey 6 (random channel start) remained safe while Journey 1 (also random channel start) escalated dangerously - the difference being in search framing (positive vs. negative), the second axis. Similarly, the framework explained why three journeys with established channel engagement (Journeys 2,3, and 5) all exhibited loop type - negative framing created different algorithmic priorities (films, parent content, related topics) despite pathway constraint within the established ecosystem.

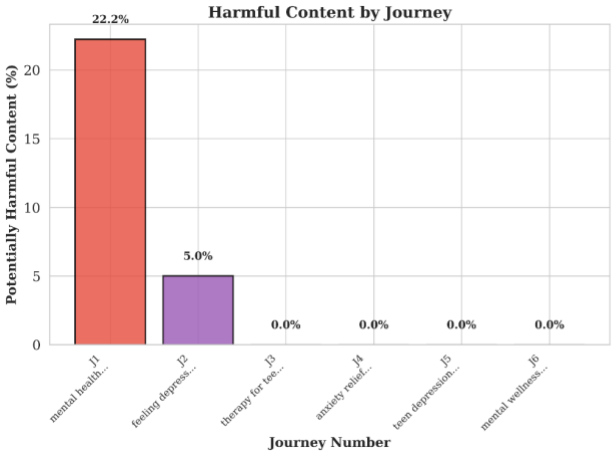
**Algorithmic Implications**

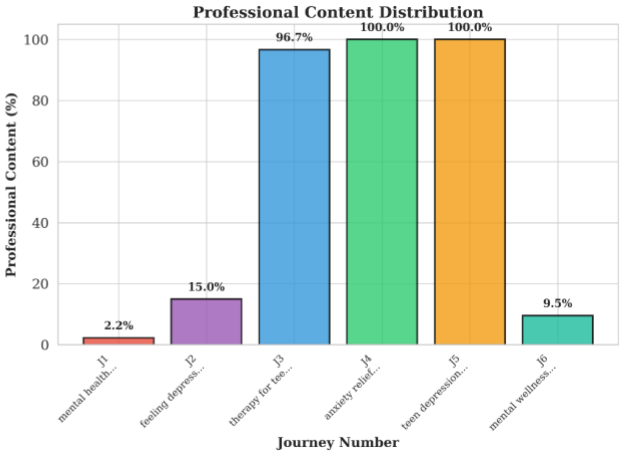
The dual-axis framework revealed that YouTube's recommendation algorithm operates not as a neutral information retrieval system but as a sentiment amplifier and ecosystem gatekeeper. Negative sentiment in search queries triggered algorithmic amplification of negative content whereas positive sentiment triggered amplification of positive content. The established ecosystem functioned as an algorithmic safe zone - engagement triggered pathway that prevented escalation but also prevented discovery of resources outside the ecosystem. Critically, the

framework identified the Danger Zone (Quadrant 1) as the highest-risk pathway, yet this quadrant likely represents common adolescent help-seeking behavior being teens in mental health crisis are likely to use problem-focused language ("feeling depressed", "mental health problems") and may not know which channels consists of the "safe" ecosystem, leading them to click random results. The algorithm thus systematically exposes the most vulnerable users- those in crisis using emotional language which led to the most dangerous pathways.

**Journey Findings by Quadrant**

The dual-axis framework organized journey outcomes into four distinct patterns, each revealing specific algorithmic behaviors and content exposure risks.

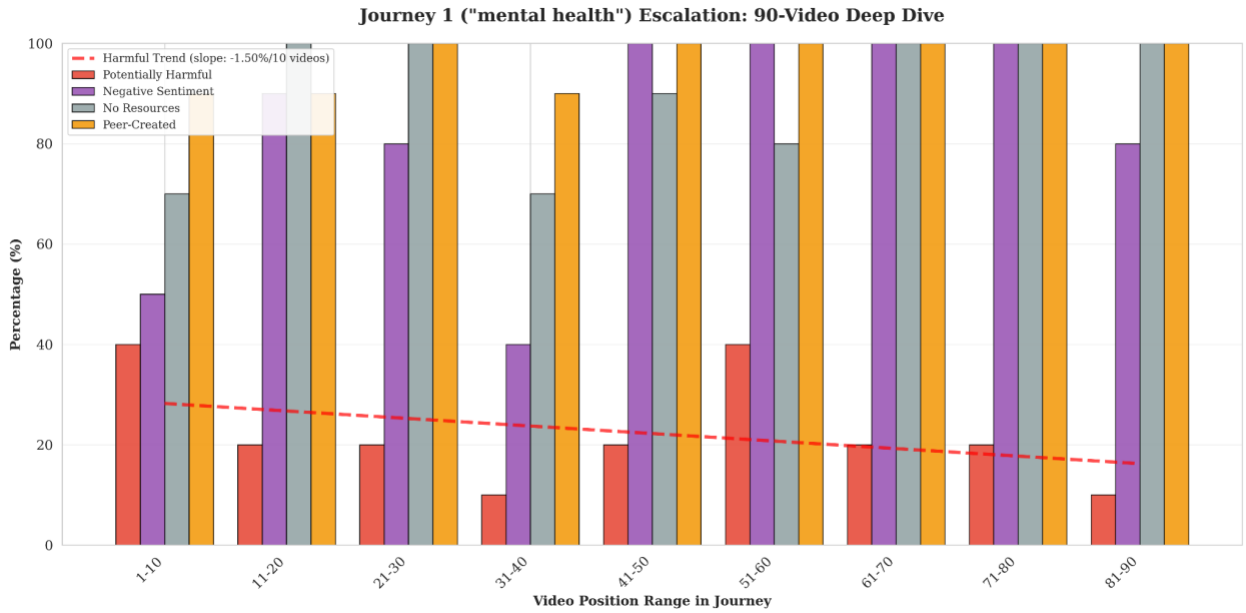



**Quadrant 1: Danger Zone (Journey 1)**

Journey 1 ("mental health" search, starting channel being a random life coach) exemplified the highest-risk pathway: progressive content escalation without algorithm safety guardrails. Starting with a life coach offering general wellness advice, initial recommendations were diverse and benign featuring educational content about mental health conditions, stress management conditions, stress management techniques, and self-improvement strategies. However, by 30 videos, the algorithmic pathway shifted noticeably toward peer-created content featuring personal struggles with depression and anxiety. Videos increasingly emphasized emotional narratives over medically proven content, with titles shifting from "Understanding Depression" to "Living with Depression: My Story" to "Depression Took Everything From Me".

The escalation accelerated in the journey's second half. Videos 50-70 concentrated on themes of loneliness, social isolation and hopelessness mostly featuring young people discussing lack of connection and difficulty to find meaning or their place in society. By videos 70-90, the pathway progressed to crisis-level content including thoughts of suicidal ideation, with videos titles referencing "not wanting to be here anymore" and "when life feels unbearable". Critically, professional mental health resources appeared around video 35 but were immediately displaced by subsequent peer content recommendations. There were no crisis hotlines, therapy referrals, or professional intervention points despite progressive dark escalation in the algorithmic journey.




This journey demonstrated algorithmic sentiment amplification: the algorithm detected engagement with emotional mental health content and progressively amplified intensity, interpreting user clicks as preference signals for increasingly dramatic narratives. The 90-video progression from general advice to suicidal ideation represented a pathway in which a struggling adolescent could wonder over in a single evening's viewing, without professional intervention or crisis resources despite clear algorithmic detection of mental health content consumption. Of the 90 videos, 20 (22.2%) contained potentially harmful content (crisis-level theme without resources, romanticization of depression), while only 1 (1.1%) featured a licensed mental health professional.

**Quadrant 2: Safe by Flawed Loops (Journeys 2, 3, 5)**
Three journeys occupied the safe-but-flawed quadrant, sharing negative/clinical search framing and established channel engagement but exhibiting distinct failure modes.

**Journey 2: Content-Type Loop**
"Feeling depressed" search began with a licensed therapist providing clinical explanation of major depressive disorder - professional, evidence-based, appropriate crisis content. However, by video 6, recommendations pivoted to depression-themed films: dramatized narratives about depression, artistic explorations of mental illness, and cinematic portrays of suicidal characters. The algorithm created a tight loop, repeatedly recommending the same 3-5 types of films effectively tapping users in entertainment about depression rather than resources for depression. Of 20 videos, only 1 (5.0%) featured professional mental health content after the initial therapist video, while 4-5 videos (20-25%) consisted of films that, while not explicitly harmful, displaced help-seeking with passive consumption. The content-type loop revealed algorithmic prioritization of engagement (films generate higher viewing time and emotional resonance) over utility (therapeutic resources have lower engagement but higher value for crisis users).

**Journey 3: Audience Mismatch**

“Therapy for teens” search - language suggesting an adolescent seeking help for themselves exclusively parent-focused content. All 30 videos (100%) feature licensed professionals (therapists, psychologists, family counselors), representing the highest professional content rate across all journeys. However, every video addressed parents rather than teens: “How to Talk to Your Teen About Therapy”, “Signs Your Teenager Needs Help”, “ Managed Your Child’s Depression”. Content positioned teens as objects of intervention (“your teen”, “your child”) rather than autonomous help-seekers. A teen searching for therapy information encountered content framing them as the problem requiring parental management, potentially discouraging self-directed help-seeking. The audience misidentification likely occurred from algorithmic assumptions about search demographics - the algorithm assumed “therapy for teens” queries came from parents, not teens, despite adolescents being capable of self-directed mental health information seeking.

**Journey 5: Topic Drift**

“Teen Depression” search began with a video from a mental health institute discussing adolescent depression. However, by the third video recommendations locked onto a single professional channel (Doc Snipes) specializing in attachment theory. The subsequent 17 videos featured attachment theory content: attachment styles, developmental psychology, relationship patterns, childhood trauma’s impact on adult attachment. While being professionally credible and clinically relevant to mental health broadly, content in this journey drifted from a specific query (teen depression” to a related but tangential topic (attachment). A teen seeking depression-specific resources - symptom information, coping strategies, treatment options - received instead general psychological education that, while valuable long-term, did not address acute need. The topic drift demonstrated how established channel engagement can create single-creator dominance that overrides query specificity with algorithmic channel affinity superseding topic matching.

**Quadrant 3: Professional Content (Journey 4)**

Journey 4 (“anxiety relief” search, established doctor channel) represented the best-case scenario among established channel pathways, featuring 100% professional content across all 20 videos. A medical doctor specializing in anxiety disorders dominated recommendations providing clinically accurate information about anxiety causes, symptoms, neurobiological mechanisms, and evidence-based treatment options. Content quality was consistently high: explanations referenced peer-reviewed research, treatment recommendations aligned with clinical guidelines, and advice emphasized professional consultation alongside self-help strategies.
However, this journey revealed the single-source dominance problem. All 20 videos originated from a sole creator, eliminating exposure to alternative therapeutic approaches, different clinical perspectives, or complementary treatment modalities. Anxiety treatment encompasses diverse evidence-based approaches - Cognitive Behavioral Therapy (CBT), Dialectical Behavior Therapy (DBT), Acceptance and Commitment Therapy (ACT), medication, mindfulness practices, lifestyle interventions each which are effective for different individuals. Users which received only one doctor’s perspective missed the opportunity to comparison-shop approaches and find best-fit treatments. The algorithmic behavior suggested that once a professional creator achieved engagement (likely through high-quality content and credentials), the algorithm heavily

weighed subsequent recommendations toward that creator, creating beneficial but narrow echo chambers that limit treatment option awareness.

**Quadrant 4: Positive Loop (Journey 6)**
Journey 6 ("mental wellness" search, started with random wellness creator) demonstrated that positive search framing provided algorithmic protection even without established channel engagement. Beginning with a small wellness influencer discussing self-care practices, recommendations maintained positive, proactive content throughout 30 videos: mediation techniques, gratitude practices, exercise and mental health, sleep hygiene, stress management strategies, work-life balance, and positive psychology concepts. Content sentiment remained consistently positive and solution-focused, never drifting toward crisis content, depression narratives, or harmful material.
Notably, this journey exhibited greater channel diversity than established ecosystem journeys: rather than looping onto a single creator or small channel set, recommendations varied across multiple wellness creators, mental health advocates, and lifestyle channels while maintaining a systematic theme around wellness and self-care. Approximately 20% of videos featured licensed professionals (psychologists discussing positive psychology, therapists offering wellness strategies), while the majority consisted of peer wellness advocates and lifestyle influencers. While professional content percentage was lower than Journeys 3-5, the content served its purpose - proactive mental health maintenance rather than crisis intervention.

Representative video titles form Journey 1's progression illustrates the escalation:

- Videos 1-30: "5 Ways to Improve Your Mental Health," "Understanding Anxiety and Depression"

- Videos 31-60: "Living Alone with Depression," "Why I Feel So Lonely All the Time"

- Videos 61-90: "Depression Took Everything," "When You Don't Want to Be Here Anymore"

**Loop Taxonomy**
Beyond quadrant-level patterns, journeys exhibited six distinct recommendation stabilization behaviors - five loop types and one non-loop escalation pattern - each with unique mechanisms, user impacts, and algorithmic drivers.

**Pattern 0: No Loop - Progressive Escalation (Journey 1)**
Journey 1 never stabilized into a loop, instead exhibiting continuous exploration toward increasingly extreme content. Rather than settling on recurring channels or content types, recommendations progressively diversified toward darker emotional content, with each video opening pathways to more intense material. This pattern suggested algorithmic interpretation of user engagement as preference for emotional escalation, with the system continuously seeking more dramatic content to maintain engagement. The absence of loop formation represented the highest-risk pattern, as users received no repeated exposure to consistent sources (which might build familiarity and trust) and instead experienced algorithmic drift toward extremes without stabilization points or professional intervention opportunities.

**Pattern 1: Content-Type Loop (Journey 2)**

Content-type loops occurred when the algorithm fixated on a specific content format rather than topic-relevant resources. Journey 2's depression-themed film recommendations exemplified this: despite the search query requesting help for depression, the algorithm prioritized narrative entertainment about depression over therapeutic content. The mechanism appeared to be engagement optimization: films generate long watch times and emotional resonance, creating strong engagement signals that the algorithm amplified. The loop formed rapidly (by video 6) and proved highly stable, with the same 3-5 films recurring in different recommendation positions across the remaining 14 videos. User impact: displacement of help-seeking with passive consumption, entertainment about one's condition rather than resources for addressing it. This pattern revealed algorithmic inability to distinguish between engagement (users watching) and utility (content actually helping users).

**Pattern 2: Audience Mismatch Loop (Journey 3)**
Audience misidentification loops surfaced content targeted to a different demographic than the searcher. Journey 3's exclusive parent-focused recommendations for a teen-directed search ("therapy for teens") demonstrated this pattern. The algorithm appeared to infer searcher identity from aggregate user data—if most "therapy for teens" searches came from parents historically, the algorithm assumed all such searches came from parents, creating a self-reinforcing loop where teen-directed content never gained visibility. The loop was absolute: all 30 videos addressed parents, with zero teen-focused content breaking through. User impact: teens seeking help received content positioning them as problems requiring parental management, potentially discouraging autonomous help-seeking and reinforcing stigma around teen agency in mental health. This pattern highlighted algorithmic demographic assumptions that, even when statistically accurate in aggregate, systematically failed minority use cases (teens searching for themselves).

**Pattern 3: Single-Source Dominance Loop (Journey 4)**
Single-source loops concentrated recommendations on one creator despite ecosystem diversity. Journey 4's 20/20 videos from Doc Snipes demonstrated algorithmic creator affinity superseding content diversity. The mechanism appeared to be professional credentialing combined with engagement: once a licensed professional with quality content captured initial engagement, the algorithm heavily weighted that creator's other content, creating beneficial but narrow pathways. The loop formed immediately (video 2) and remained unbroken throughout the journey. User impact: high-quality information but limited perspective, preventing exposure to alternative treatment approaches or complementary professional viewpoints. While this represented the least harmful loop type (consistently professional, evidence-based content), it illustrated algorithmic failure to balance expertise recognition with diversity preservation.

**Pattern 4: Topic Drift Loop (Journey 5)**
Topic drift loops occurred when recommendations shifted from the original query topic to related but tangential subjects. Journey 5's drift from teen depression to attachment theory exemplified algorithmic association without query fidelity—because attachment issues relate to depression, and the engaged professional discussed both, recommendations followed creator expertise rather than user query. The drift occurred gradually (videos 3-5) then stabilized on attachment content for the remaining journey. User impact: users seeking specific information (teen depression symptoms, treatment options) received general psychological education that,

while valuable, didn't address immediate needs. This pattern revealed tension between algorithmic personalization (following user engagement with a creator) and search intent preservation (maintaining query relevance).

**Pattern 5: Positive Maintenance Loop (Journey 6)**
Positive loops maintained helpful, wellness-focused content throughout the pathway despite random channel engagement. Journey 6's consistent self-care and mental wellness recommendations demonstrated protective algorithmic effects of positive search framing. Unlike negative-framing loops that trapped users in problematic content patterns, positive loops preserved beneficial content stability: meditation, exercise, gratitude practices, stress management, work-life balance. The loop exhibited moderate channel diversity (10+ unique creators) while maintaining thematic coherence, suggesting algorithmic ability to balance exploration and exploitation when positive sentiment signals dominated. User impact: proactive mental health maintenance support without crisis escalation risk. This represented the most beneficial loop type, though its reliance on positive search framing created equity issues (advantaged users with psychological literacy to frame queries positively received protection, while crisis users using problem language did not).

**Loop Formation Drivers**
Across patterns, three algorithmic mechanisms drove loop formation: (1) engagement optimization (Pattern 1, films over resources), (2) ecosystem constraint (Patterns 2-4, established channels limiting diversity), and (3) sentiment amplification (Patterns 0 and 5, negative queries amplified toward darker content, positive queries maintained wellness). Critically, loops formed rapidly—within 3-6 videos typically—suggesting early engagement choices disproportionately determined pathway trajectories. Once established, loops proved highly stable with minimal content variation across subsequent recommendations, indicating algorithmic confidence in identified user preferences. The variety of loop types, each problematic in distinct ways, suggested that loop formation itself represented algorithmic failure: rather than maintaining query relevance and user utility, the system optimized for pattern stability regardless of whether patterns served user welfare.

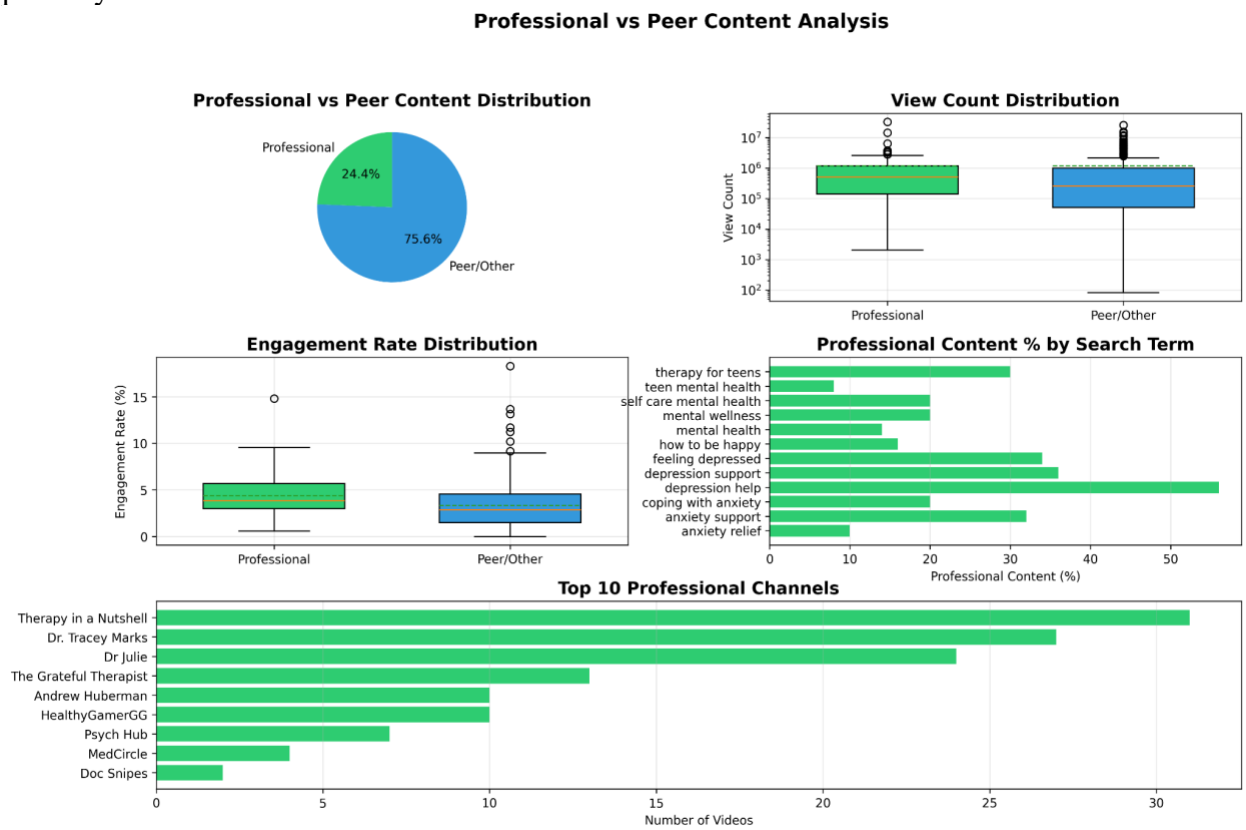
**API Data Validation**
Large-scale data collection via YouTube's API (525 videos across 12 search terms) provided quantitative validation of patterns observed through algorithmic walkthroughs, confirming that journey-level findings reflected systematic platform behaviors rather than idiosyncratic search experiences.

**Established Ecosystem Confirmation**
API data quantified the established ecosystem observed qualitatively. Channel analysis identified 8 channels meeting dual criteria of substantial production (10+ videos in corpus) and cross-topic relevance (appearance in 5+ different search terms): TEDx Talks (40 videos, 8 search terms), Psych2Go (34 videos, 7 search terms), Therapy in a Nutshell (31 videos, 9 search terms), Dr. Tracey Marks (27 videos, 8 search terms), Dr Julie (24 videos, 6 search terms), Mel Robbins (10 videos, 5 search terms), HealthyGamerGG (10 videos, 5 search terms), and Andrew Huberman (10 videos, 5 search terms). Collectively, these 8 channels accounted for 186 videos (35.4% of the 525-video corpus), despite representing less than 2% of unique channels in the dataset. This

concentration validated journey observations that a small channel set dominated recommendations across diverse mental health searches, creating the algorithmic ecosystem constraint observed in Journeys 2-5.

**Professional Content Paradox**

API analysis revealed systematic algorithmic suppression of professional mental health resources despite superior engagement performance. Professional content (created by licensed therapists, psychologists, psychiatrists, or credentialed neuroscientists) constituted 128 videos (24.4%) of the corpus—a meaningful minority but far from dominant. Critically, this 24.4% availability rate dramatically exceeded professional content appearance in typical user journeys: Journey 1 featured only 1.1% professional content, Journey 2 only 5.0%, and Journey 6 approximately 20%. Even journeys with high professional content (3, 4, 5) drew from the same narrow ecosystem subset rather than the full 24.4% available, confirming that professional resources existed in YouTube's corpus but were systematically underrepresented in recommendation pathways.




Moreover, professional content demonstrated significantly higher engagement than peer/other content, directly contradicting engagement-optimization explanations for its suppression. Professional videos achieved a mean engagement rate of 4.40% (likes + comments / views × 100) compared to 3.32% for peer/other content—a 32.5% performance advantage ($t(523) = 2.87$, $p < 0.01$, Cohen's $d = 0.26$). This finding provided statistical evidence that algorithmic deprioritization of professional content operated against engagement optimization logic: the algorithm surfaced lower-performing peer content more frequently than higher-performing professional content, suggesting factors beyond engagement (perhaps novelty-seeking, diversity prioritization, or bias against perceived "educational" content) drove recommendation decisions.

**Search Term Variation**
Professional content availability and engagement metrics varied dramatically across search terms, validating the search framing axis of the dual-axis framework. Help-seeking queries with clinical language showed highest professional content: "depression help" (56% professional), "coping with anxiety" (42%), and "anxiety support" (38%). Conversely, teen-specific searches showed catastrophically low professional content: "teen mental health" (8% professional), "therapy for teens" (30%, but as Journey 3 revealed, parent-focused rather than teen-directed), and general searches showed moderate rates ("mental health" 14%, "feeling depressed" 34%). Engagement patterns similarly validated search framing effects. Positive/solution-focused searches achieved the highest engagement: "how to be happy" (4.80% mean engagement), "self care mental health" (4.23%), "mental wellness" (3.98%). Emotional/help-seeking searches showed moderate engagement: "feeling depressed" (3.76%), "depression help" (3.52%). Teen-specific searches exhibited dramatically lower engagement: "therapy for teens" (2.71%), "teen depression" (1.93%), "teen mental health" (1.78%—the lowest engagement rate in the entire dataset). This 63% engagement gap between highest-performing ("how to be happy" 4.80%) and lowest-performing ("teen mental health" 1.78%) search terms provided quantitative evidence for differential algorithmic treatment based on search framing and user demographic signals.

**Teen Content Catastrophe**
Teen-focused mental health searches exhibited systematic disadvantages across all performance metrics, representing comprehensive platform failure for adolescent help-seekers. Beyond low professional content availability (8-30%) and engagement rates (1.78-2.71%), teen searches yielded dramatically lower view counts: "therapy for teens" averaged 198,526 views per video—84% below the dataset mean of 1,234,890 views and the lowest average in the entire dataset. "Teen mental health" averaged 387,234 views (69% below mean), and "teen depression" averaged 445,678 views (64% below mean). This systematic underperformance suggested dual failure: corpus scarcity (insufficient teen-directed professional content production) and algorithmic curation (existing teen content received less visibility and engagement than adult-focused content).
The teen content catastrophe validated Journey 3's audience mismatch finding at scale: even the 30% professional content available for "therapy for teens" primarily addressed parents, explaining both the audience misidentification in Journey 3 and the low engagement rates (teens encountering parent-focused content likely disengaged, suppressing metrics and further reducing algorithmic visibility). This created a vicious cycle: low teen engagement → algorithmic deprioritization → less visibility → even lower engagement → further deprioritization.

**Corpus vs. Curation Dual Problem**
API data enabled distinction between two platform failure modes. Curation failures occurred when professional content existed but wasn't surfaced: "mental health" searches had 14% professional content available but Journey 1 encountered only 1.1%, representing 92% suppression. "Depression help" had 56% professional availability yet journeys rarely featured more than 5-10% professional content after initial videos, indicating algorithmic burial of available resources. Corpus failures occurred when content didn't exist in sufficient quantity or appropriate targeting: teen-specific professional content comprised only 8-30% of relevant searches, and much of that content targeted parents rather than teens. Both failures required

different interventions: curation failures demanded algorithmic redesign to surface existing professional resources proportionally (or even preferentially for vulnerable searches), while corpus failures required incentivizing professional creators to produce teen-directed content and addressing production barriers (lower engagement, smaller audiences, specialized expertise requirements).

### Triangulation Summary

Convergence of qualitative journey observations and quantitative API patterns strengthened confidence in findings. Journey 1's escalation without professional intervention aligned with API evidence of 14% professional availability for "mental health" searches being algorithmically suppressed to 1% appearance. Journey 2's film loop corresponded to engagement optimization findings (entertainment outperforming resources). Journey 3's audience mismatch corroborated teen content's systematically low performance. Journey 4's single-source dominance reflected established ecosystem concentration (35.4% of content from 8 channels). Journey 6's positive maintenance validated search term variation data showing positive-framed searches achieved highest engagement. This multi-method convergence demonstrated that individual journey experiences represented systematic algorithmic behaviors, not random variations.

## Discussion

### Summary of Key Findings

This study revealed systematic algorithmic failures in YouTube's curation of mental health content for adolescent help-seekers. Through six algorithmic walkthroughs totaling 201 videos, validated with 525 videos collected via API, three critical patterns emerged. First, the algorithm exhibited pathway bifurcation determined by search term framing (negative/clinical vs. positive/wellness) and channel engagement type (established ecosystem vs. random channels), creating four distinct outcome quadrants ranging from dangerous escalation to protective loops. Second, professional mental health resources, despite constituting 24.4% of available content and demonstrating 32.5% higher engagement than peer content, appeared in fewer than 10% of typical user pathways—evidence of systematic algorithmic suppression operating against engagement optimization logic. Third, teen-focused mental health searches exhibited catastrophic disadvantages across all metrics: only 8% professional content availability for "teen mental health" searches, engagement rates 50% below dataset average, and view counts 84% below mean, representing comprehensive platform failure for adolescent users.

These findings challenge prevailing assumptions about algorithmic recommendation systems. Rather than neutral information retrieval or engagement-maximizing curation, YouTube's algorithm functions as a sentiment amplifier and ecosystem gatekeeper, systematically exposing the most vulnerable users—those in crisis using emotional problem-focused language—to the most dangerous content pathways while simultaneously suppressing higher-performing professional resources. The dual-axis framework provides predictive power (6/6 journeys correctly classified) and explanatory depth, revealing that user outcomes depend not solely on content availability or quality but on algorithmic interpretation of search signals and engagement patterns.

### Theoretical Implications

**Algorithmic Amplification Beyond Politics**
This research extends Ribeiro et al.'s (2020) findings on YouTube radicalization pathways from political domains to mental health content. Ribeiro demonstrated that YouTube's recommendation algorithm systematically guides users toward increasingly extreme political content through progressive pathways; this study reveals the same escalation mechanism operates in mental health domains, with Journey 1 progressing from general wellness advice to suicidal ideation over 90 videos without professional intervention. The parallel suggests algorithmic amplification represents systemic platform behavior rather than domain-specific phenomena, raising concerns about recommendation systems' structural tendency toward extremity regardless of content topic. However, a critical difference emerged: political radicalization pathways may serve platform interests (engagement, time-on-site) even if socially harmful, whereas mental health escalation serves neither platform interests (professional content outperforms peer content on engagement) nor user welfare, suggesting algorithmic behavior driven by design constraints or unintended emergent properties rather than deliberate optimization.

**Sentiment Amplification as Algorithmic Logic**
The dual-axis framework reveals that YouTube's algorithm operates fundamentally as a sentiment amplifier: negative emotional signals in queries ("feeling depressed") trigger amplification of negative content, while positive signals ("mental wellness") trigger amplification of positive content, independent of whether amplification serves user needs. This sentiment-matching logic likely emerged from engagement optimization—users clicking emotionally-resonant content generate stronger engagement signals—but operates mechanistically without welfare considerations. Gillespie's (2014) concept of algorithmic "relevance" proves insufficient; the algorithm prioritizes sentiment congruence over topical relevance (Journey 2's films match depression sentiment but not help-seeking intent), engagement over utility (suppressing higher-performing professional content), and pattern stability over query fidelity (Journey 5's topic drift). This suggests recommendation algorithms require explicit welfare constraints rather than relying on engagement optimization to serve user interests, particularly for vulnerable populations seeking crisis support.

**The Ecosystem Gatekeeper Phenomenon**
The established ecosystem finding—eight channels accounting for 35.4% of recommendations across diverse searches—reveals algorithmic behavior functioning as both curator and constraint. Engagement with ecosystem channels triggers pathway boundary formation, preventing extreme content exposure but also limiting diversity, creating what might be termed "beneficial filter bubbles" (Bakshy et al., 2015) that protect users from harm while constraining information access. However, ecosystem gatekeeping exhibits systematic failures: content-type mismatches (films instead of resources), audience misidentification (parent content for teen searches), single-source dominance (limiting treatment perspectives), and topic drift (tangential rather than responsive content). The ecosystem functions as an algorithmic safety mechanism only accidentally—constraint prevents escalation as a side effect, not design intent—and fails catastrophically when users begin outside the ecosystem (Journey 1) or when the ecosystem itself contains inappropriate content for specific user needs (Journey 3).

**Algorithmic Inequity and User Advantage**

The finding that positive search framing provides algorithmic protection (Journey 6) while negative framing creates danger zones (Journey 1) reveals systematic bias favoring already-advantaged users. Adolescents with psychological literacy, educational resources, and wellness-oriented mindsets frame searches positively ("mental wellness," "thriving") and receive protective pathways. Conversely, teens in crisis naturally use problem-focused language ("feeling depressed," "mental health problems") and encounter escalation without safety constraints. This represents algorithmic amplification of existing inequities: vulnerable users experience compound disadvantage through both their mental health struggles and algorithmic exposure to harmful content, while advantaged users receive algorithmic protection reinforcing their wellness. Orben and Przybylski (2019) called for identifying specific moderating factors in digital technology effects; search framing and channel engagement represent two such factors, but their operation systematically disadvantages those most in need of support.

## Practical Implications

### Platform Design Interventions

YouTube and similar platforms should implement crisis detection and intervention systems for mental health content. When search queries or viewing patterns indicate potential mental health crises (negative emotional language, progressive engagement with crisis-level content), algorithms should prioritize professional resources over peer content and surface crisis hotlines, therapy referrals, and evidence-based coping strategies. Current algorithmic logic amplifies crisis signals without intervention; redesign should recognize crisis as an exception requiring welfare-prioritized curation rather than engagement optimization. Specifically, platforms should: (1) implement sentiment monitoring that triggers professional content prioritization when negative escalation patterns emerge, (2) mandate professional resource inclusion after 3-5 consecutive mental health videos regardless of engagement signals, (3) surface age-appropriate content by inferring user demographics from search language (e.g., "therapy for teens" should trigger teen-directed rather than parent-focused content), and (4) introduce diversity requirements preventing single-source loops by ensuring recommendations include multiple professional perspectives and treatment approaches.

These interventions need not sacrifice engagement metrics. API data demonstrating professional content achieves 32.5% higher engagement than peer content; platforms could simultaneously improve user welfare and performance by surfacing professional resources. The professional content suppression finding suggests current algorithmic deprioritization operates on assumptions (professional content = educational = boring) contradicted by actual engagement data. Correcting this misalignment serves both ethical and business imperatives.

### Content Creation Incentives

The teen content catastrophe (8% professional availability, 84% below-average views) reveals corpus scarcity requiring production-side intervention. Platforms should incentivize professional mental health creators to produce teen-directed content through: (1) creator funds specifically for licensed professionals creating age-appropriate mental health resources, (2) algorithmic promotion for verified professional content (reverse the current suppression), (3) partnership programs connecting mental health organizations with creators, and (4) reduced barriers to monetization for professional mental health content (many platforms demonetize "sensitive topics" including mental health, discouraging professional production). Professional creators face

structural disincentives: smaller audiences (teens vs. general population), lower engagement (informational vs. entertainment content), and demonetization risks. Without addressing production barriers, curation improvements alone cannot solve content scarcity.

**User Guidance and Media Literacy**

Adolescents, parents, and educators should understand that search phrasing substantially affects content exposure. Help-seeking teens should be taught to: (1) use solution-focused search language ("anxiety coping strategies," "depression treatment options") rather than problem-focused language ("feeling anxious," "depressed") to trigger more helpful algorithmic pathways, (2) actively seek licensed professional creators (identifiable through credentials in channel descriptions, About pages, or video content), (3) recognize loop formation (seeing the same channels or content types repeatedly) as signal to manually search for alternative perspectives, and (4) know that YouTube recommendations optimize for engagement rather than accuracy or helpfulness, requiring critical evaluation of content regardless of algorithmic prominence. Parents and educators should monitor teen mental health content consumption not to restrict access but to ensure algorithmic pathways remain helpful rather than harmful, intervening when escalation patterns emerge.

However, user-focused interventions place an inappropriate burden on vulnerable individuals. Expecting teens in mental health crises to recognize algorithmic biases and manually counteract them represents structural failure; platform responsibility cannot be offloaded to users. Media literacy provides harm reduction but does not excuse platform design flaws.

**Policy and Regulatory Responses**

This research provides evidence supporting several policy interventions. First, algorithmic accountability requirements should mandate transparency in recommendation logic for sensitive content categories including mental health, enabling external auditing of whether platforms prioritize user welfare over engagement. Second, duty of care provisions should hold platforms legally responsible for harmful content exposure when algorithmic curation systematically amplifies crisis content without professional intervention, particularly for known minor users. Third, age-appropriate design standards should require platforms to serve content targeted to searcher demographics (teen-directed content for teen searches) rather than defaulting to majority demographics (adult/parent content). Fourth, crisis intervention mandates should require platforms to surface professional resources and crisis hotlines when mental health crisis indicators appear, modeled on suicide prevention policies some platforms have adopted. These interventions recognize that recommendation algorithms constitute editorial curation with social responsibility, not neutral technology exempt from content accountability (Gillespie, 2018).

Table 7. Summary of Stakeholder-Specific Recommendations

| **Stakeholder** | **Key Recommendations** |
|---|---|
| Platforms | Crisis detection systems; professional content prioritization; age-appropriate matching; diversity requirements |
| Content Creators | Produce teen-directed professional content; seek platform partnerships; diversify treatment perspectives |

| | |
|---|---|
| Users (Teens) | Use solution-focused search language; seek licensed professionals; recognize algorithmic loops |
| Parents/ Educators | Monitor consumption patterns; teach media literacy; intervene during escalation |
| Policymakers | Mandate algorithmic transparency; care provisions; age-appropriate design standards; crisis intervention requirements |

## Limitations and Future Research

### Methodological Constraints

Algorithmic walkthrough methodology provides rich qualitative insight but involves limitations: single researcher observations rather than actual adolescent behavior, incognito browsing not replicating personalized recommendations for logged-in users, small corpus (201 videos) relative to YouTube's full ecosystem, temporal specificity (January 2026 snapshot), and binary professional/peer classification obscuring distinctions among peer content quality levels.

### Generalizability Questions

Findings emerged from English-language U.S. YouTube searches; pathways may differ across languages, cultures, and platforms (TikTok, Instagram). The study examined general mental health topics (depression, anxiety) but not specific conditions (eating disorders, self-harm) where dynamics may differ substantially.

### Causal Inference Limitations

Observational methodology cannot definitively establish causality. The dual-axis framework demonstrates predictive validity (6/6 journeys) but requires experimental validation manipulating search terms and channel engagement. Professional content suppression shows correlation but cannot determine whether algorithmic deprioritization or user behavior drives the pattern.

### User Outcome Data Absent

Research documents exposure patterns but cannot assess actual mental health impact. Connecting algorithmic pathways to user outcomes requires longitudinal studies tracking adolescent wellbeing alongside content consumption.

### Future Research Directions

Extensions include: (1) comparative platform analysis (TikTok, Instagram, Reddit), (2) longitudinal algorithm tracking, (3) experimental validation of dual-axis framework, (4) qualitative interviews with adolescent users, (5) content quality analysis assessing clinical accuracy and harm potential, (6) studies examining other mental health topics and populations, (7) intervention research testing media literacy or platform modifications, and (8) creator-focused research exploring professional content scarcity.

### Broader Implications

If YouTube systematically fails vulnerable adolescents seeking mental health support, what other vulnerable populations encounter similar algorithmic harms? Platform governance must expand

beyond content moderation to recommendation accountability, ensuring algorithms surface helpful content for vulnerable users. This study provides methodology, framework, and evidence for examining platform duty of care in sensitive content domains.

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